

Computational Social Science

Barış Arı

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Essential Digital Tools

These tools form a **basic power-user toolkit** for working with data, code, and research projects.

1. Code Editors

Use a professional code editor instead of basic text editors.

VS Code

VS Code has become the industry standard in both commercial and academic environments.

It is extremely powerful and integrates easily with many essential tools used in modern research and development, including:

- **AI coding assistants**
- **Git and GitHub**
- an integrated terminal and development environments

VS Code provides syntax highlighting, extensions, integrated terminals, debugging tools, and support for many programming languages. This makes it a flexible environment for writing code, managing projects, and running analysis workflows in one place.

You can use it for pretty much any language you like, including Python and R.

Website: <https://code.visualstudio.com/>

Sublime Text

Many users also keep **Sublime Text** installed. Sublime is a very lightweight editor that starts instantly and is useful for quickly opening and editing files. Because of its speed and simplicity, many users keep both editors installed and switch between VS Code and Sublime depending on the task.

Website: <https://www.sublimetext.com/>

2. Git and GitHub

Git is a **version control system** that records the history of your project.

Core ideas:

- track changes in files
- create saved versions of a project (**commits**)
- collaborate with others

We cover this in class.

3. Command Line / Terminal

The terminal allows you to **interact directly with the operating system**.

Typical uses:

- navigate folders
- run programs
- manage files
- use development tools

Windows and macOS/Linux use slightly different terminal environments (for example **PowerShell** or **Command Prompt** on Windows, and **Terminal/Bash** on macOS and Linux). However, the basic logic is the same: you run commands that tell the computer what to do.

The goal is not to memorise every command, but to develop a **basic understanding of how the system works** and how to move around your files and run tools from the terminal.

While learning the basics, AI tools can help explain commands and common errors.

You can access the terminal through VS Code.

For Windows, you can check this guide: <https://www.digitalcitizen.life/command-prompt-how-use-basic-commands/>

4. Understanding File Locations

Students must understand **where their files actually live on the computer**.

Key ideas:

- files are organised in **directories (folders)**
- programs run in a **working directory**

- paths describe file locations

Essential skill:

- navigate directories easily in the terminal

A guide for Unix-like systems (MacOS, Linux, etc.), but also relevant for Windows users:
<https://carpentry.library.ucsb.edu/2022-07-12-ucsb-bash/02-filedir/>

Note that Windows and Unix-like systems organise files in a very similar way. The main visible difference is the **path separator**:

- Windows uses \
- Unix systems (Linux, macOS) use /

5. Markdown

Markdown is a **simple markup language** for writing formatted documents.

This very document is written in Markdown.

Uses:

- documentation
- reports
- notes
- reproducible research

Markdown has become a widely used standard for technical writing and documentation. The basics are very simple and can be learned quickly. Some useful tutorials:

- <https://learn-markdown.github.io/>
- <https://www.markdownguide.org/>
- <https://www.markdowntutorial.com/>

Related tools:

- **Quarto** (optional): <https://quarto.org/docs/get-started/hello/rstudio.html>
- **LaTeX** (optional): https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes

6. Virtual Environments and Reproducibility

Research projects often depend on specific software packages and versions.

Virtual environments allow you to:

- isolate project dependencies
- avoid conflicts between projects
- reproduce results reliably